

I started my first programming class and my first Latin class on the same day, as a nervous high school freshman. I quickly fell in love with both classics and computing, and I began to see the profound connections between computation and language. Diagramming Latin sentences was remarkably similar to tracing the control flow of a Java program, and the skills I learned from breaking down real-world program requirements into classes and functions also helped me break down the complex grammar of Cicero and Caesar. I dreaded starting college because I would have to choose a major, and deciding between my two passions felt impossible. As a college freshman, I started working for the Digital Latin Library. My first project at DLL was writing scripts to scrape Latin text from web pages, associate the text with its metadata, and write the results to a database. A few months into this work, I watched as the Latin text of the Vulgate Bible filled my terminal window. Watching my code work was as thrilling as the knowledge that I had done something novel and unique. I knew at that moment that I wanted a career in research at the intersection of technology and language. I declared dual majors in computer science and letters (interdisciplinary humanities) and started exploring research at the intersection of my two passions.

My desire to work on complex technical problems pushed me toward deep learning, and my connection to the humanities encouraged me to specialize in natural language processing (NLP), especially machine translation (MT) because of the untapped potential of applying MT to literature. Studying classics had shown me how much human understanding of the world is shaped by narrative and myth; digital humanities taught me that our perception of our own past and present is largely determined by who had the education to write and the resources to share their writing. Reading someone's story, as told in their own words, is one of the most powerful ways to understand another person, and large-scale MT of literary texts can give marginalized people a global platform to tell their stories. Based on this goal, my primary research interest is developing improved methods for neural machine translation of extremely low-resource (LR) languages, for which little to no parallel training data is available. These languages are largely spoken by historically marginalized communities, including indigenous people and those living in developing countries. My research focuses on low-resource MT because it is both a difficult, theoretically interesting NLP task and an opportunity to mitigate social inequality by improving access to existing resources for immigration, education, healthcare, and other issues. I am especially interested in building MT systems capable of translating literary works, with the result having literary merit comparable to a human translation. Literary MT is an unsolved problem even for high-resource languages; combined with the difficulties of low-resource translation, this is the grand challenge to which I plan to devote my career.

Intellectual Merit/Relevant Background: NLP is a fundamentally interdisciplinary field, requiring programming skills, strong machine learning background, and domain knowledge of linguistics. I am uniquely qualified to conduct this work because of my dual academic background. Like many of my peers, my CS and math classes have prepared me for the technical aspects of NLP. Unlike my peers, however, I have a strong background in morphology and syntax from studying Latin, meaning I see language as a complex, intricately structured system to be explored, rather than just a dataset on which to train deep learning models. I also have a liberal arts education that enables me to explain computational problems to collaborators with a variety of technical and nontechnical backgrounds, as well as years of practical experience working with linguistic corpora.

Throughout my undergraduate career, I was a research assistant for the Digital Latin Library. DLL is a collaborative project between several universities and academic professional societies whose purpose is to publish open-access digital critical editions of Latin texts, many of which are foundational to modern disciplines like history, literature, philosophy, astronomy, and mathematics. At DLL, I developed scripts that made it possible for classicists with little to no technical background to publish their work in XML. XML digital editions allow readers to use interactive reading environments on the text to see variations between manuscripts more easily than in a printed edition. On average, using my scripts improves XML encoding time by a factor of about 400 compared to manual encoding by an expert human. In addition to writing and maintaining code for the DLL, I gave three presentations at national conferences and wrote my undergraduate honors thesis on evaluating what constitutes scholarship in the digital humanities. In this text-encoding work, there are always a few complicated edge cases where the

editor's argument about the text cannot be encoded with a deterministic algorithm, and defining the boundary between automatable and non-automatable tasks is an active area of research in digital humanities.

Through my work for the DLL, I developed many skills that will benefit me as an NLP researcher. Because I frequently presented technical papers at classics conferences, I excel at communicating computer science concepts to audiences from a variety of disciplines. This also allows me to write accessible and understandable papers, so that both experts and non-experts can understand the importance of my research. Through creating digital editions that other researchers can use for NLP projects, I have developed a practical knowledge of how to design and maintain linguistic corpora. I had the opportunity to further refine this knowledge at the summer 2018 NEH Institute for Advanced Topics in Digital Humanities. I was the only computer scientist, and the only undergraduate, at the Institute. This gave me the opportunity to watch how scholars from various disciplines interact with current NLP tools and where problems tend to arise. My background in digital humanities and building digital editions will help me develop NLP tools and methods benefitting a variety of users.

Between graduating from college and starting graduate school, I was a postbaccalaureate research intern at Los Alamos National Laboratory. I worked in the Ultrascale Systems Research Center, and my project focused on using NLP and machine learning to analyze supercomputer system logs. I chose this internship for two reasons: first, I enjoy learning about high-performance computing (HPC) systems almost as much as I love NLP; second, modern neural MT is highly dependent on HPC, and I knew the experience would make me a better MT researcher. I had several notable accomplishments during my internship, including generating a large dataset of syslog data from scratch. Because I was working remotely due to the COVID-19 pandemic, security rules did not allow me to access syslogs from actual LANL systems. Therefore, I spent the first half of my summer generating 750,000 lines of non-secure syslog data. In the second half of the summer, I experimented with various word embedding algorithms for extracting predictive features from the syslog data. Ultimately, I found that NLP techniques for feature extraction did not perform well on their own, but could be useful as an augmentation for other failure prediction models.

During my LANL internship, I learned important technical and nontechnical skills that are serving me well as a first-year PhD student. First, after twelve weeks working on an HPC project, I am more knowledgeable about HPC than the average NLP student, which is helpful when planning my research and allocating computational resources to different jobs. I also developed my independence and self-direction during my internship. As an undergraduate researcher, I was largely writing code to implement my supervisors' ideas. At LANL, however, I learned to take ownership and lead my own project by reading the relevant literature, developing ideas for potential methods, and then presenting my own ideas to my mentor for her input. I also improved at planning my research work by setting small, intermediate goals. The skills I learned at LANL are helping me stay productive and organized while balancing an RA appointment, graduate coursework, diversity programs, and my own research. Finally, my LANL internship gave me the opportunity to write and submit my first scientific paper. It was accepted to the Second Workshop on Machine Learning for Computing Systems at Supercomputing 2020. I hope to return to LANL for another internship during my PhD, using what I have learned in my graduate NLP coursework to build on my previous work.

Now, as a PhD student at University of Southern California Information Sciences Institute (ISI), my primary focus is MT for Central American indigenous languages, with the eventual goal of providing translation software to organizations that offer legal assistance to refugee speakers of these languages. During the first year of this work, I have established a baseline translation result for Mixtec, and I am currently working on compiling data and establishing a baseline for K'iche'. I am also working on domain adaptation in both high and low resource settings to further improve MT performance. My future plans for this work are discussed in detail in my Graduate Research Plan. USC and ISI have all of the resources necessary to carry out the research in my proposal. In particular, I have access to computational resources necessary to train large MT models as well as a strong academic community in the ISI Natural Language Group. Outside of ISI, USC has several programs focused on immigration and human rights, including

the Gould School of Law Immigration Clinic, the Keck Human Rights Clinic, and USC Shoah Foundation. My advisor has existing collaborations with these organizations and with faculty in the Dworak-Peck School of Social Work. As a USC ISI student, I have access to resources and collaborators necessary to accomplish the work described in my Graduate Research Plan.

Broader Impacts: Outside of research, I want to contribute to the field of computer science by helping increase diversity in STEM. I have been successful in this field because I had excellent mentors who shared my identities and appreciated how those identities inform my scientific perspective. I have dedicated significant time in both college and graduate school to mentoring younger women and queer people in engineering. I also work to create a supportive intellectual community for my mentees and peers by serving as a leader in student organizations for women and LGBTQ+ STEM students.

My first meaningful experience with diversity efforts was in Alpha Sigma Kappa, a sorority for women in technical studies. As soon as I joined, my new sisters passed my résumé to recruiters, introduced me to their mentors, and helped me apply to campus leadership roles. A few years later, as chapter vice president, I recruited thirty-two new members and oversaw their member education and integration into the chapter. I am proud to say that many of those women are now serving in chapter leadership. The following year, as chapter president, I led my chapter's sudden transition to virtual operations due to COVID-19. Three days after we received news that our university would be going online indefinitely, my executive board had a plan for every remaining event of the semester, as well as a schedule for virtual study sessions to support our mission of academic excellence, even during distance learning.

While I was VP, I took on the role of Committee Chair for Halliburton Women's Welcome out of necessity. The program's founder left the university for another job, and a new director was hired three weeks before HWW. In her first meeting with the planning committee, she was brutally honest: she had no idea how to execute this program. Looking around the room, I realized I was the only person with any previous HWW experience and that, if I wanted our students to have the same empowering experience I had as a participant, I was going to have to take charge and execute the program. Over the next few weeks, I supervised other committee members while also doing my official committee duties of organizing a campus resource fair and developing an inclusive professional dress seminar, which received 91% positive feedback on the post-program participant survey. At the end of the week, 80 young women were excited to start engineering degrees, felt confident about their ability to succeed, and knew that they had plenty of mentors ready to support them.

Currently, I am in my second year as Graduate Representative for USC QuEST (Queers in Engineering, Science, and Technology). In this role, I both recruit current graduate students to QuEST and mentor QuEST undergraduates during their graduate school applications. During my first term, I increased the number of graduate students active in QuEST, tripled the number of graduate students serving on QuEST leadership, and gave a presentation on graduate school and fellowship applications for our undergraduate members. In March, I was invited to give a presentation on LGBTQ+ allyship for student leaders at USC's annual Women in Engineering Conference. Additionally, I am currently serving as Outreach Chair on the USC WiE advisory board, where I am responsible for collaboration with other student organizations and K-12 outreach events. I look forward to mentoring women in STEM throughout my graduate studies and eventually as a faculty member. I want to pay forward the investment that all of my mentors have made in me by serving my community in the same way.

Future Goals: After my PhD, my career goal is to be a professor of computer science at an R1 institution. I have enjoyed my experiences tutoring both college and K-12 mathematics, and I have seen how supportive teachers can help students who were previously afraid of math and computer science learn to be excited about them. In my research career, I will continue to improve low-resource MT, especially for literary text. MT currently faces limitations in many areas, including sentiment, connotation, and referential cohesion (determining what nouns actually refer to the same entity and translating appropriately). I am particularly interested in how morphologically complex languages like Latin can be used to improve existing MT systems. I will use my background in computer science, classics, and interdisciplinary research to further advance the field of machine translation.